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AI/ML-Enhanced Wind Forecasts for Reducing Uncertainty in Prescribed Fire Planning

Sara Brambilla · S. Coffing · J. Slaten · Diego Rojas · D. Robinson · Arvind Mohan

Environmental Science, Computer Science · Atmosphere · 18 March 2026

TLDR By providing high-resolution, bias-corrected, and probabilistic wind fields, AI/ML-enhanced forecasts will allow for expanded burn windows, improved ignition strategy design and a reduced reliance on expert intuition, especially when a prescribed **fire** is introduced into new areas. Expand

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AI Methods Tailored to Influenza, RSV, HIV, and SARS-CoV-2: A Focused Review

Achilleas Livieratos · G. Kagadis · Charalambos Gogos · K. Akinosoglou

Medicine, Computer Science · Pathogens · 30 July 2025

TLDR Standardized open-access data pipelines and integration of explainable-AI methodologies are recommended to ensure safe, equitable deployment of AI-driven interventions in future viral-outbreak responses. Expand

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Advances in the Automated Identification of Individual Tree Species: A Systematic Review of Drone- and AI-Based Methods in Forest Environments

Ricardo Abreu-Dias · Juan M. Santos-Gago · F. Martín-Rodríguez · Luis M. Álvarez-Sabucedo

Environmental Science, Computer Science · Technologies · 6 May 2025

TLDR This systematic **review** explores the latest research on drone-based data collection and AI-driven classification **techniques**, focusing on studies that classify specific tree species rather than generic tree detection, and reveals that **deep learning** (DL) models are increasingly replacing traditional ML methods such as random **forest** or support vector **machines**. Expand

5 Publisher Save Cite

When AI Meets Cache: Intelligent Memory Hierarchy Management in Modern CPUs

Yihan Wang · Suchuan Xing · Computer Science, Engineering · Computer Life · 25 March 2026

TLDR This **review surveys** the state of the art at the intersection of AI and cache **management**, examining **techniques** including **deep** reinforcement **learning**, long short-term memory (LSTM) networks, transformer-based sequence models, and graph neural networks as applied to the three principal problem domains of cache replacement, hardware prefetching, and adaptive resource partitioning. Expand

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AI-BASED PHISHING WEBSITE DETECTION

Ravi Kumar Mahone · Nishanshu Deshmukh · Nishant Sahu · Sahil Darokar · Sangam jain

Computer Science · International Journal of Leading Research… · 14 November 2025

TLDR This **review** paper explores the transition to **Artificial Intelligence** and **Machine Learning** as a robust, adaptive solution to phishing, tracing the evolution from simple ML **techniques** to state-of-the-art **Deep Learning** and hybrid systems. Expand

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AI for Automated Code Reviews and Quality Assurance

Mariappan Ayyarrappan · Computer Science · International Scientific Journal of Engineering… · 16 November 2024

TLDR How AI **techniques**, such as language models and **machine learning**–based code analysis, can enhance automated code **reviews** and ensure consistent standards at scale is discussed. Expand

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AI-Assisted Theorem Discovery: Revolutionizing Mathematical Exploration

Yuli Bangun Nursanti · Imam Sujadi · I. Kurniawati · A. Wulandari · Riki Andriatna

Mathematics, Computer Science · 2025 International Conference on Frontier… · 27 March 2025

TLDR A systematic **review** of the different types of **machine learning techniques** that have been utilized for student dropout prediction, examining their approaches, datasets, metrics, and practical use cases. Expand

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AI-Based Water Usage Monitoring and Reduction in Hospital Operations

Prof. Prashant Sharma · Environmental Science, Medicine · International Journal of Latest Technology in… · 29 June 2025

TLDR By adopting AI-driven approaches, hospitals can not only reduce operational costs but also contribute to broader sustainability goals, including compliance with national green building standards and global climate targets. Expand

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Analysing Multidimensional Strategies for Cyber Threat Detection in Security Monitoring

P. Shelke · Timo Hamalainen · Computer Science, Engineering · European Conference on Cyber Warfare and Security · 21 June 2024

TLDR A comprehensive overview of recent research into novel methods for cyber threat detection, encompassing diverse approaches such as **machine learning**, **artificial intelligence**, behavioral analysis and anomaly detection, providing a concise yet comprehensive guide to modern cyber threat detection strategies. Expand

8 PDF PDF Save Cite

An Overview of Different Approaches used for Processing Satellite Images

C. V. · Rashmi S · Environmental Science, Computer Science · 2024 International Conference on Distributed… · 20 September 2024

TLDR A comparative study **survey** on various approaches used for processing of satellite images using **deep learning**, **Artificial intelligence** and **machine learning** to get fast and more accurate results is provided. Expand

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